Spatial AI Model Comparison for Building Detection

Area of Interest: Purchase, NY

Julie Amon, Carlyce McIntosh,
Kimberly McNeil, Spring 2026

PROJECT GOAL/STUDY AREA

- **Our goal:**
  Train different segmentation models on the same AOI dataset and compare how well they detect buildings.

- **Area of Interest:** Residential area within Purchase, New York. This clipped raster section was used to train and evaluate building detection models.

DATASET DESCRIPTION

Raster Imagery: Clipped TIFF image of a residential area within Purchase, NY

Ground Truth Labels: Building footprint polygons stored as a GeoJSON file

The models were trained to identify buildings from the raster imagery, using the building footprint labels as the correct answer for comparison.

Fig. 1 Input Raster

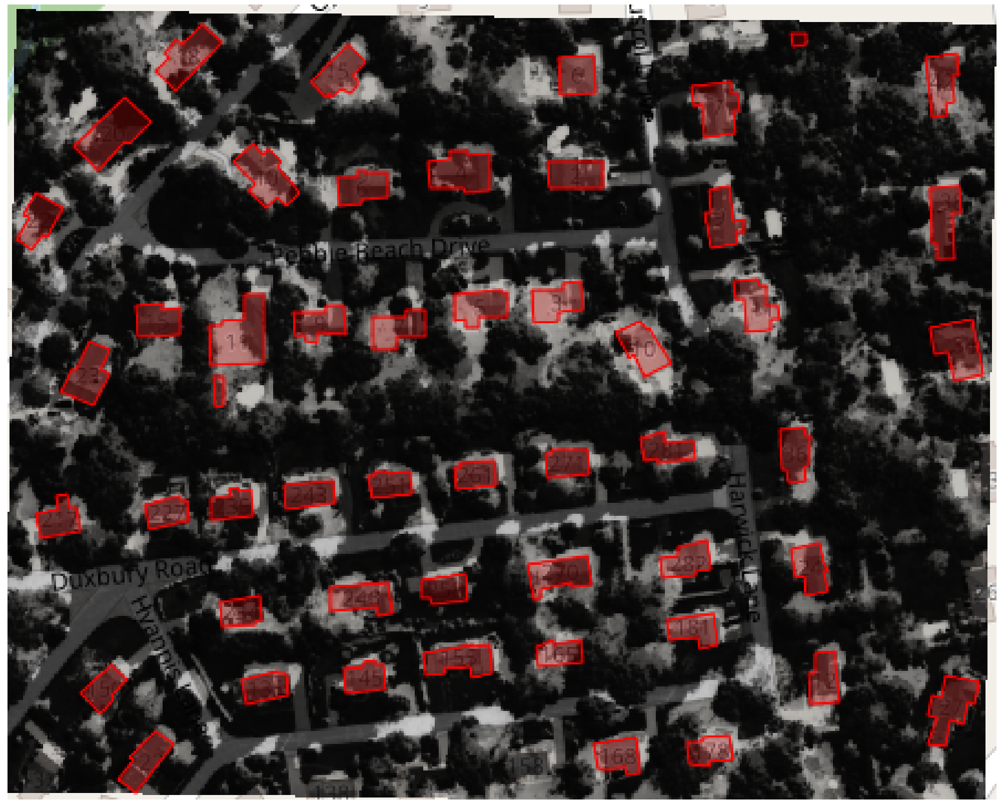


Fig. 2 Ground Truth Labels

MODELS USED

U-Net

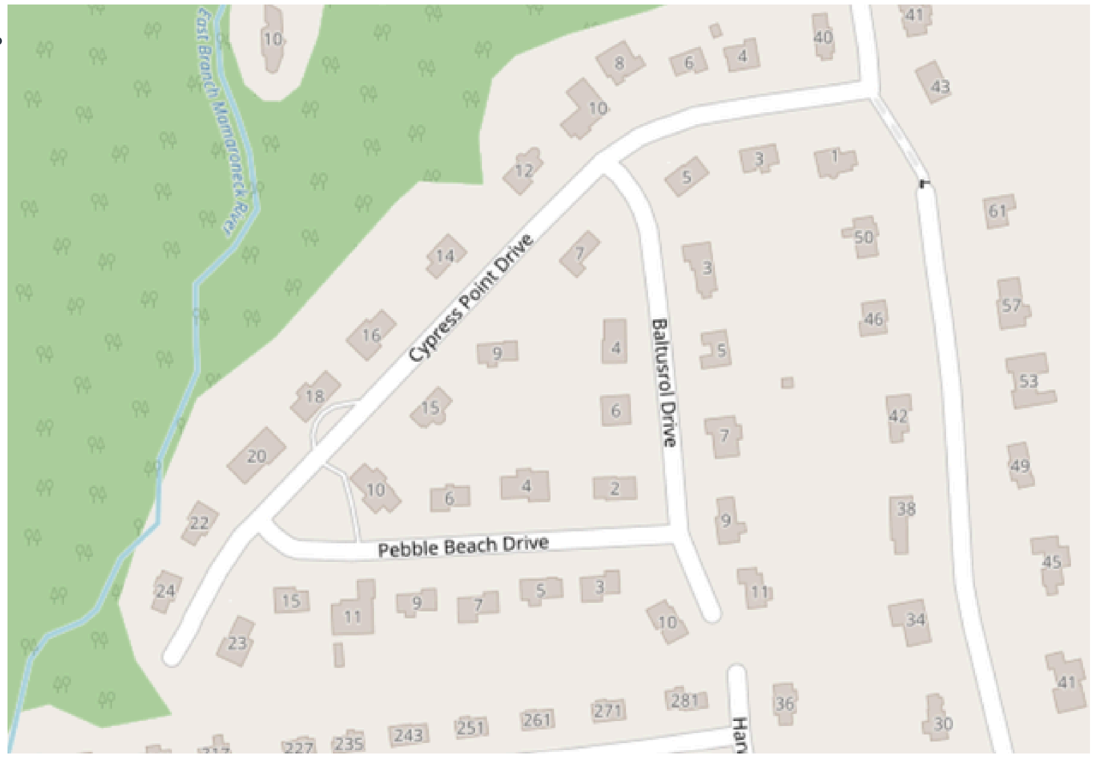
Convolutional Neural Network (CNN)-based segmentation model used as a baseline for detecting buildings from raster imagery.

DeepLabV3

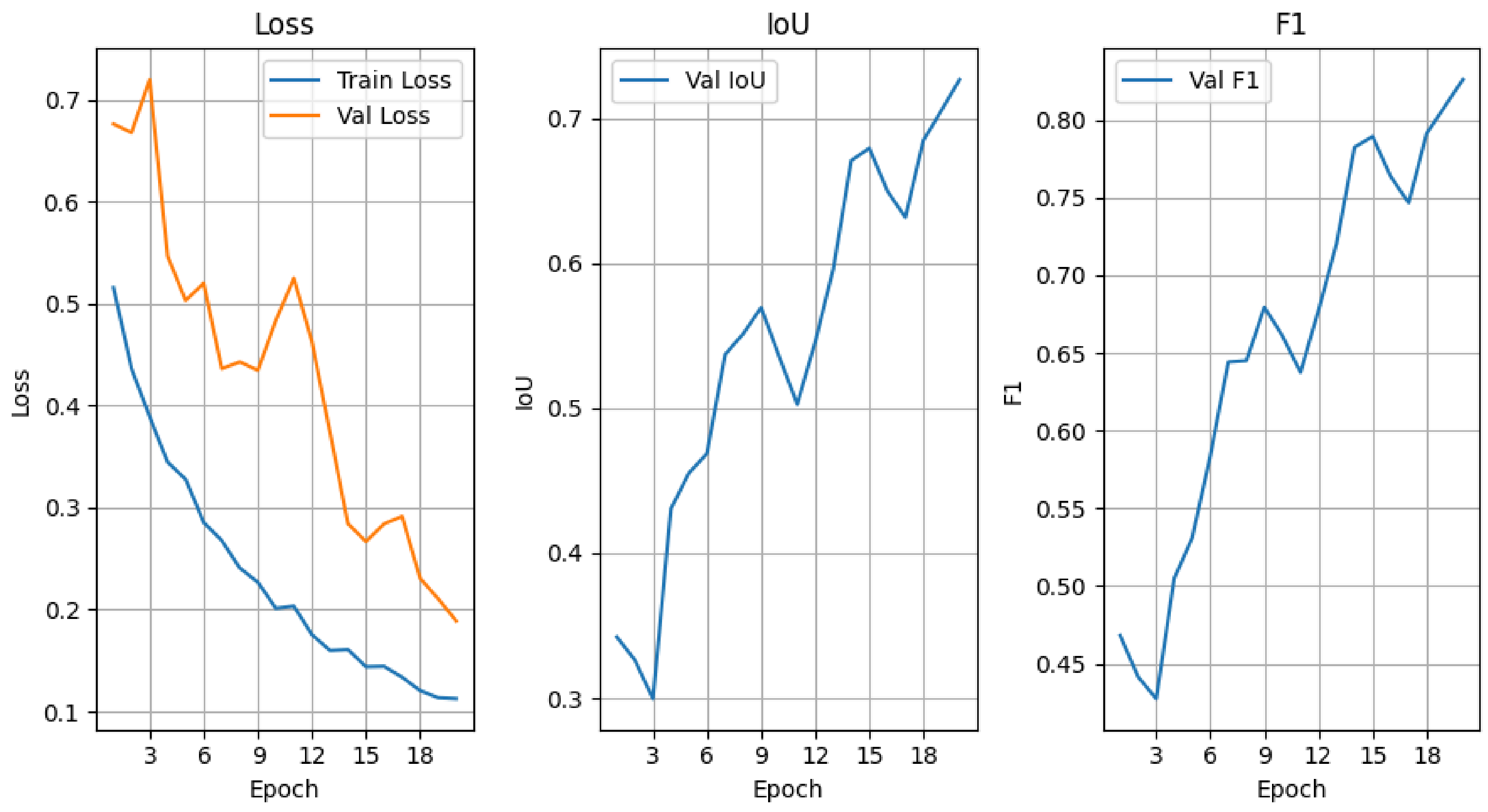
Segmentation model that uses spatial context to identify objects at different scales.

SegFormer

Transformer-based segmentation model designed to capture broader image patterns and relationships.

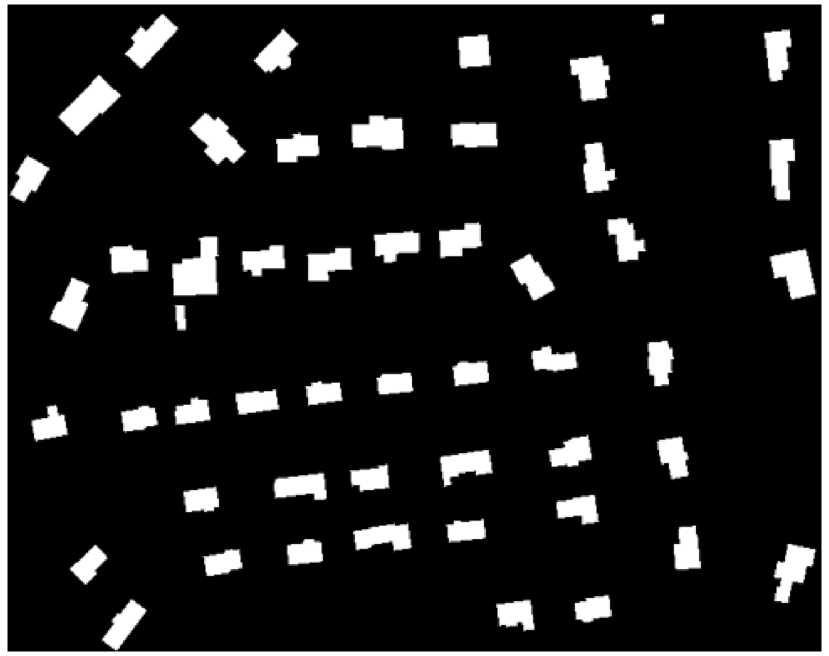


U-NET

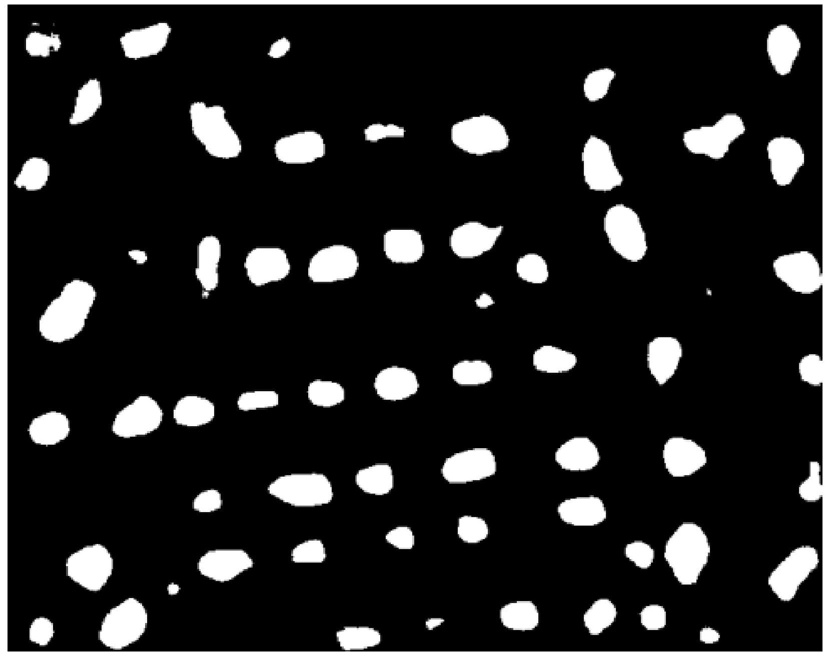


	epoch	train_loss	val_loss	val_iou	val_f1	val_precision	val_recall
0	1	0.515585	0.676102	0.342247	0.468228	0.524386	0.562153
1	2	0.436116	0.667489	0.326273	0.441332	0.496096	0.490267
17	18	0.120493	0.230606	0.684806	0.791298	0.778925	0.805413
18	19	0.113582	0.210638	0.705454	0.808623	0.800336	0.817597
19	20	0.112655	0.188739	0.726722	0.825923	0.821308	0.830728

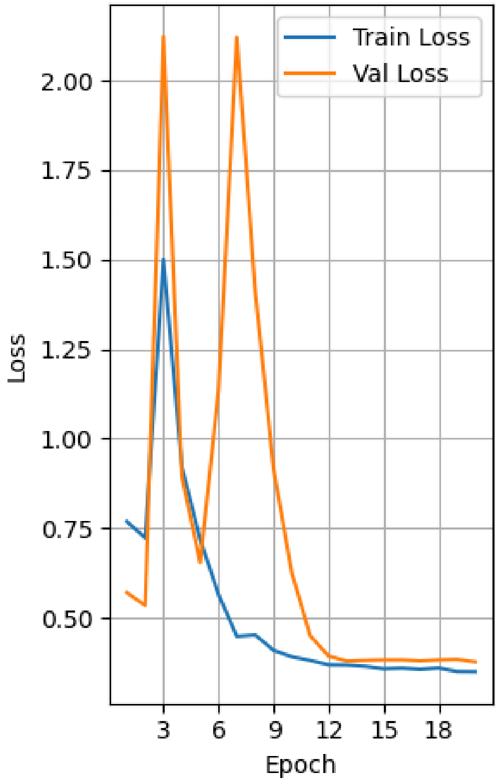
Ground Truth Labels

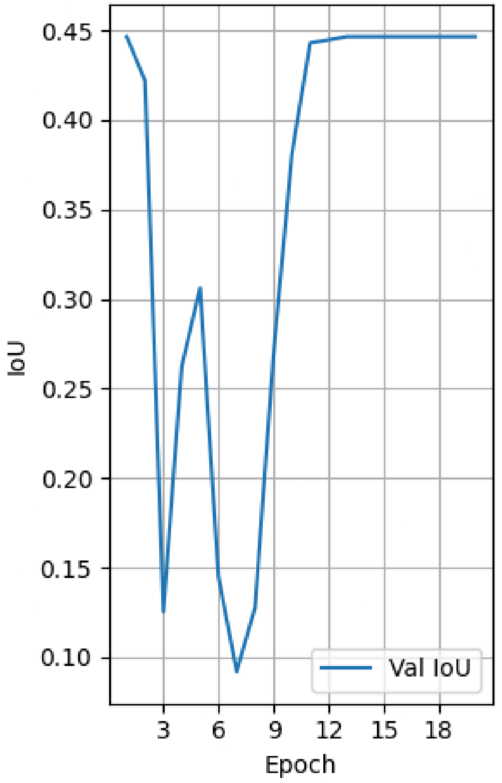


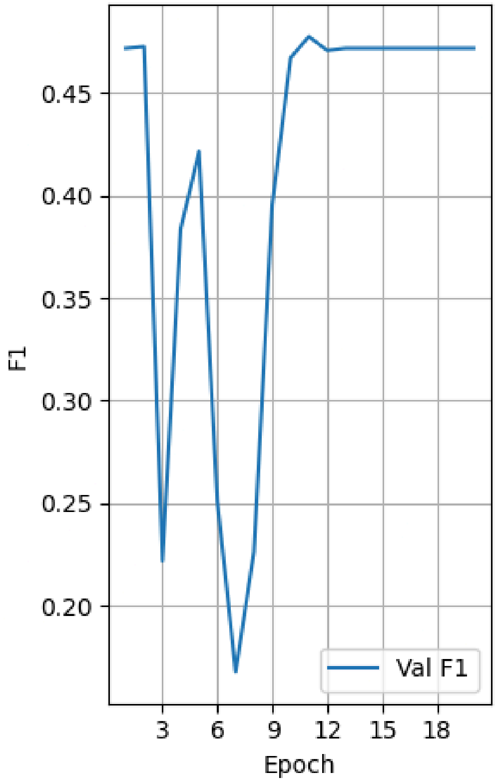
U-Net Predicted Mask



DEEPLABV3

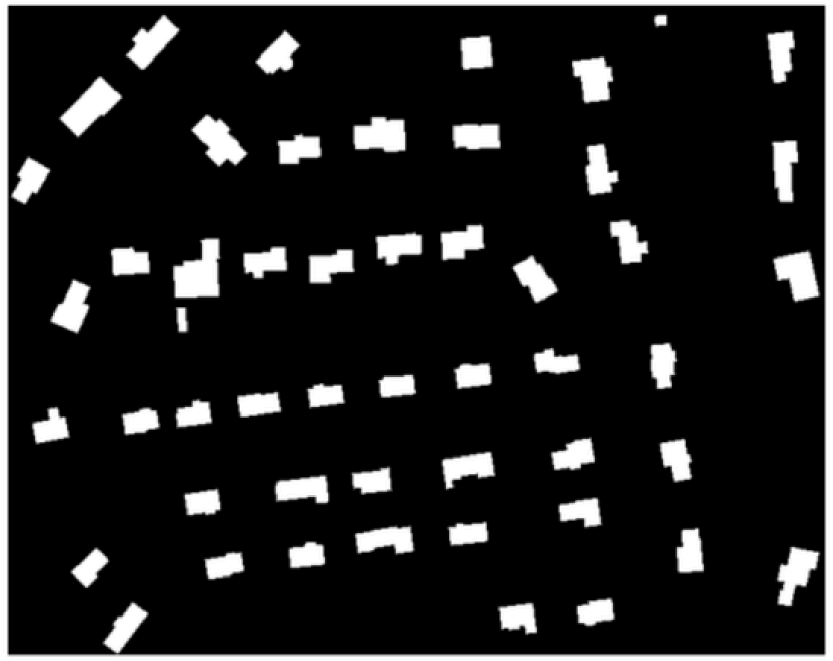




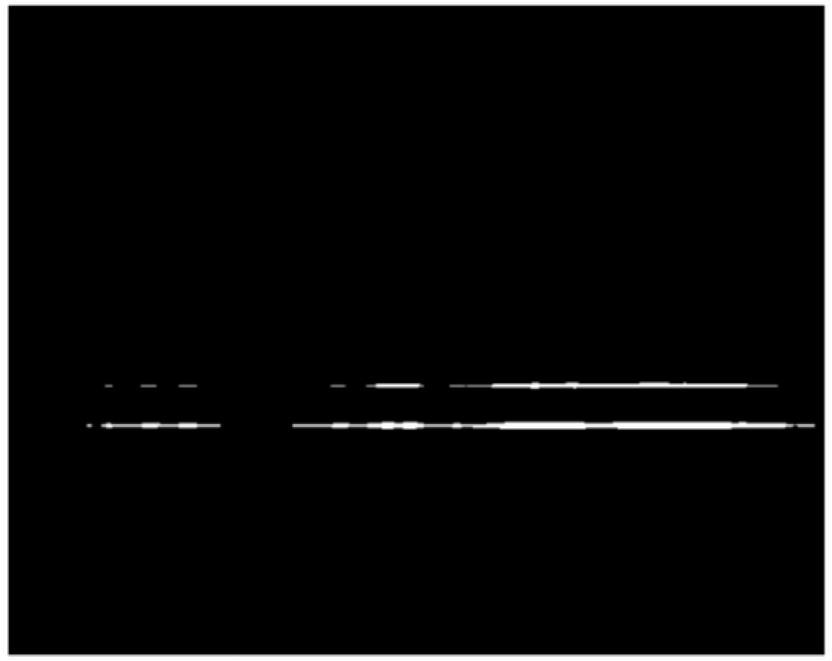


	epoch	train_loss	val_loss	val_iou	val_f1	val_precision	val_recall
0	1	1.286362	0.613139	0.379489	0.508175	0.547026	0.616580
1	2	1.237539	7.140954	0.053423	0.096532	0.106846	0.500000
17	18	0.359362	0.382134	0.446577	0.471781	0.893154	0.500000
18	19	0.349526	0.383629	0.446577	0.471781	0.893154	0.500000
19	20	0.348950	0.376308	0.446577	0.471781	0.893154	0.500000

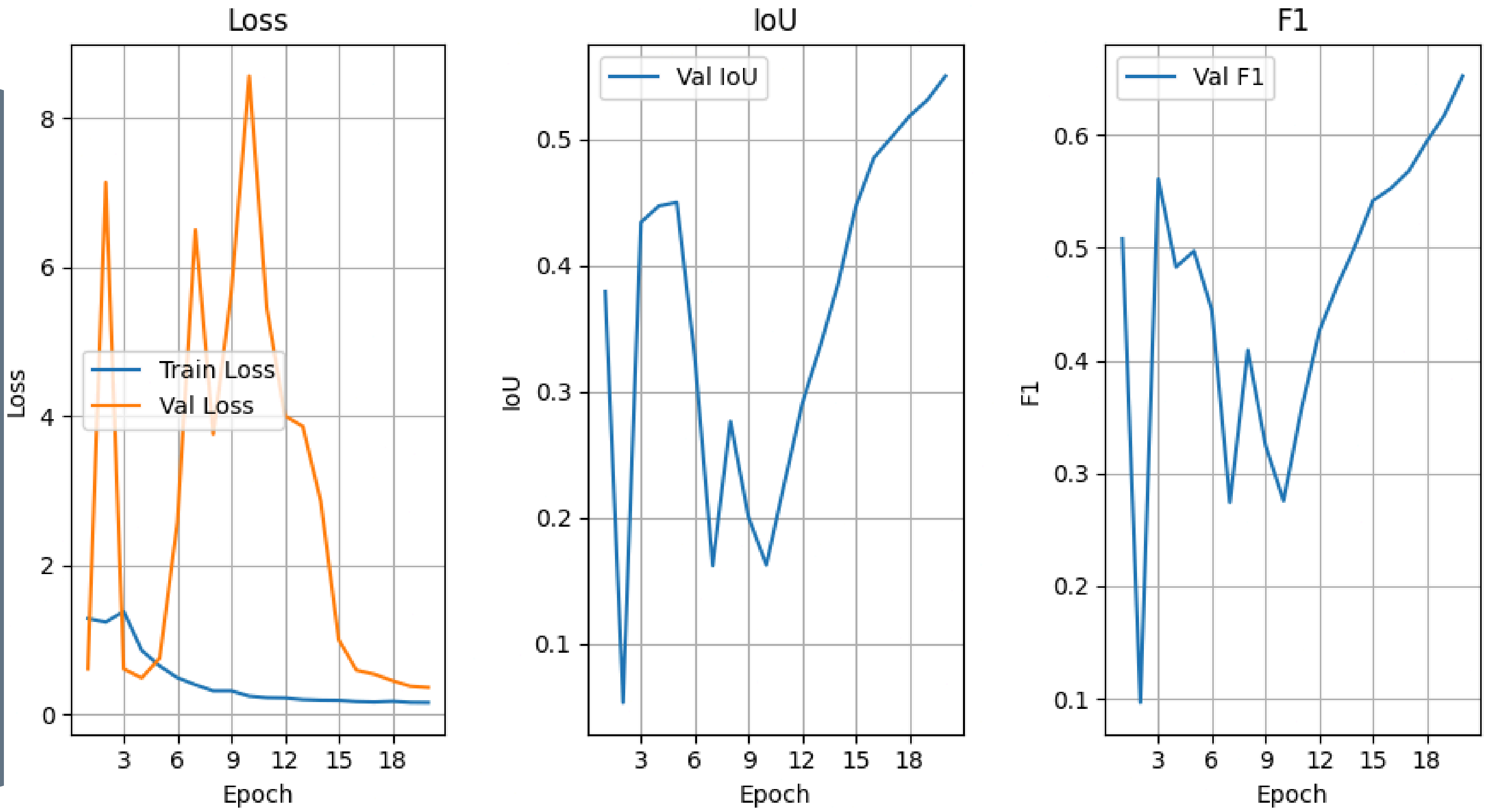
Ground Truth Labels



DeepLabV3 Predicted Mask



SEGFORMER



	epoch	train_loss	val_loss	val_iou	val_f1	val_precision	val_recall
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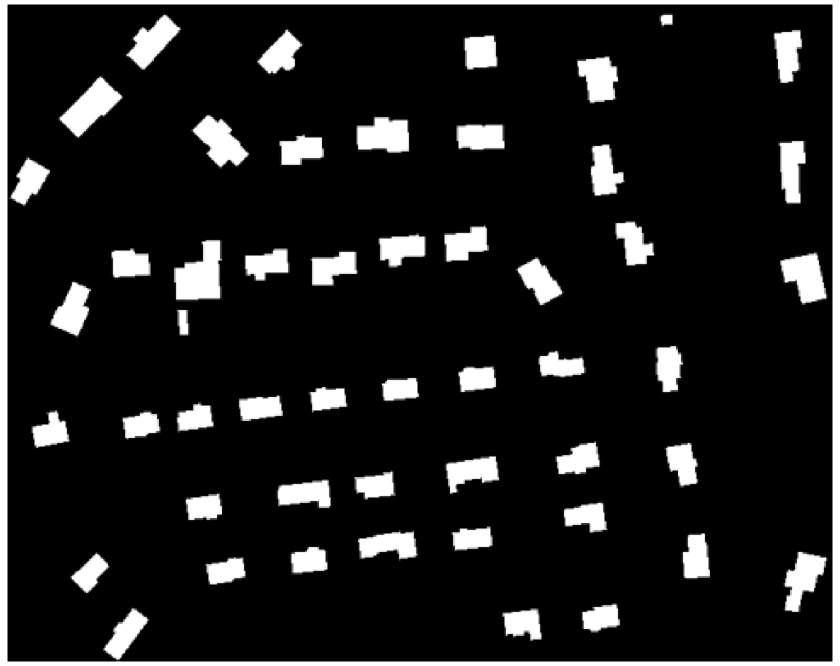
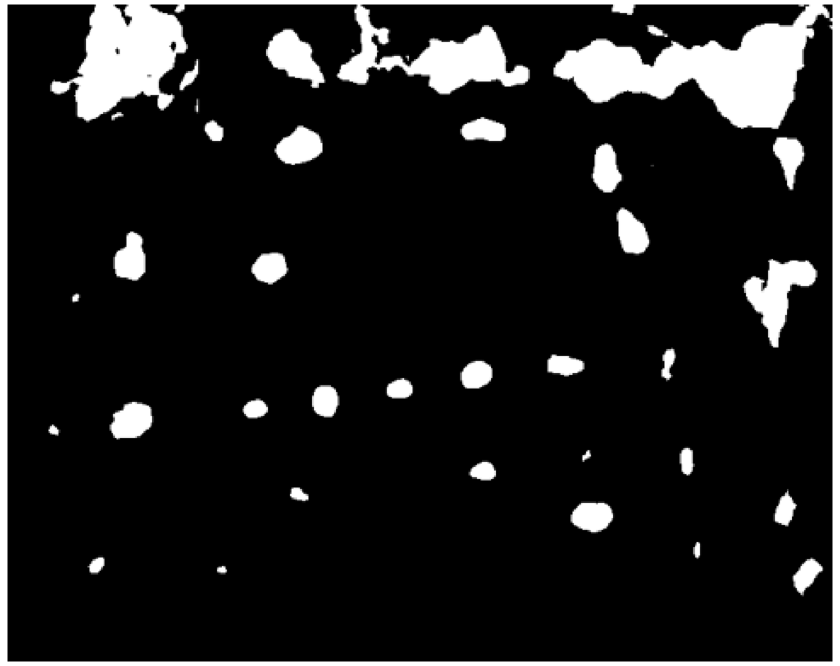
1	2	1.237539	7.140954	0.053423	0.096532	0.106846	0.500000
17	18	0.175288	0.453539	0.518603	0.594399	0.793094	0.569280
18	19	0.160390	0.378304	0.531442	0.618161	0.703608	0.591924
19	20	0.158738	0.361837	0.550251	0.652201	0.663975	0.642760

Ground Truth Labels



Segformer Predicted Mask



COMPARISON

U-Net

IoU: 0.726702
Dice: 0.825923

Succeeds: High precision in smaller datasets and boundary delineation

Fails: long ranges and large-scale/complex structures.

DeepLabV3

IoU: 0.446577
Dice: 0.471781

Succeeds: Multi-scale object segmentation and capturing wide context

Fails: fine-grained edge details, high computational resources

SegFormer

IoU: 0.550251
Dice: 0.652201

Succeeds: global dependency modeling and real-time inference (good for edge devices)

Fails: can require more training data and potentially lower edge precision

INTERPRETATION

- Which model performed best and why?
  U-Net
  - Strongest qualitative results (U-Net had the highest IoU and Dice Score)
  - Best visual match to the ground truth labels
- Poor data can lead to poor results. The dataset quality can affect performance with
  - Inaccurate or inconsistent labeling
  - Lack of diversity
  - Noise or artifacts
  - Insufficient data

CONCLUSION

Thank You